Physics 5C Discussion (Week 3)

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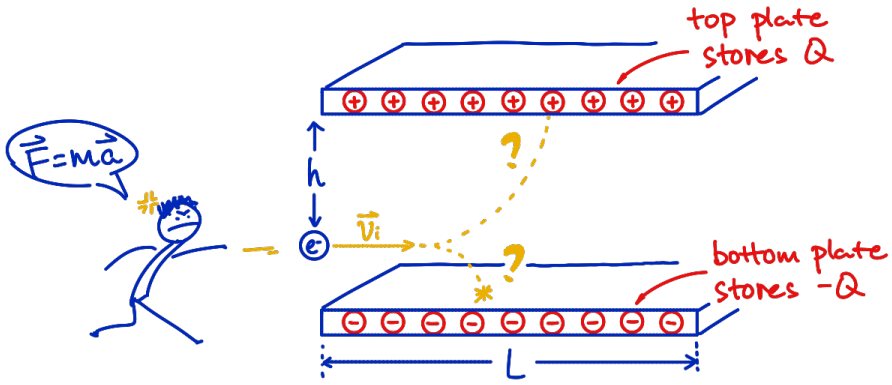
Problem 1: Consider a parallel-plate capacitor, formed by two conducting plates that carry charges as shown. The electric field in between the plates is uniform (being the same everywhere, in both direction and magnitude). The magnitude of the electric field between the plates is

$$|\vec{E}| = \frac{Q}{\epsilon_0 A},$$

with A being the area of each of the plates and Q being the amount of positive charge stored at the top plate.



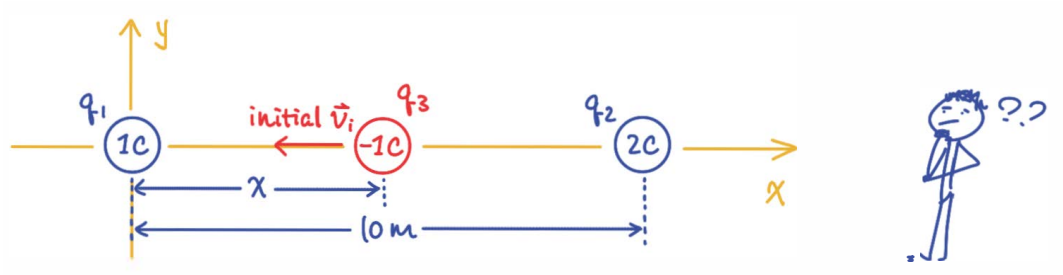
Part (a) Based on the sign of the charges stored at the two plates, determine the direction of the electric field, and write down the vector form of the electric field $\vec{E} = (E_x, E_y)$ between the plates in terms of the variables Q, A and the constant ϵ_0.

Part (b) Now Jinyan throws an electron with charge q into the region between the two plates. The electron experiences a force due to the electric field. Compute the electric force that the electron experiences, write it in its vector form $\vec{F} = (F_x, F_y)$ in terms of the unknown variables q, Q, A and the constant ϵ_0.

Part (c) The electron (with mass m) enters the region with initial velocity $\vec{v}_i = (v_{0,x}, 0)$, at a position h meters below the top plate. Each of the plates is L meters long. Symbolically compute the minimal speed $v_{0,x}$ needed such that the electron can pass through the region without hitting either plate. Neglect gravity and friction.[1]

[1] Hint: This part is a Physics-5A type of problem, you can solve it using either conservation of energy (if acceleration $\vec{a}$ is not a constant) or kinematic equations (if $\vec{a}$ is a constant). Your answer should be in terms of q, Q, m, L, h, A and ϵ_0.

Problem 2: Now we revisit the problem on the week-2 worksheet again.



$q_1 = 1\,\mathrm{C}$ is fixed at $\vec{x}_1 = (0, 0)\,\mathrm{m}$.
$q_2 = 2\,\mathrm{C}$ is fixed at $\vec{x}_2 = (10, 0)\,\mathrm{m}$.
$q_3 = -1\,\mathrm{C}$ is initially between q_1 and q_2 on the x-axis with some initial velocity.
The Coulomb constant $k = 8.99 \cdot 10^9\,\mathrm{Nm^2/C^2}$.

Part (a) Find the electric potential in the region between the two positive charges (that is, you neglect the negative charge in the middle, for this part).

Part (b) Find the electric potential energy of the negative charge q_3, when it is placed at $\vec{x} = (x, 0)$ between q_1 and q_2.

Part (c) If q_3 is initially at $\vec{x} = (5, 0)\,\mathrm{m}$, with initial velocity $\vec{v} = (-1, 0)\,\mathrm{m/s}$. Will it be able to pass through the point where it experiences zero net force? [2]

[2] Hint: From the week-1 worksheet, q_3 experiences zero net force at $x = 4.14\,\mathrm{m}$. Apply energy conservation to see whether q_3 has enough initial kinetic energy to overcome the potential energy difference between the starting location at $x = 5\,\mathrm{m}$ and the target location at $x = 4.14\,\mathrm{m}$.